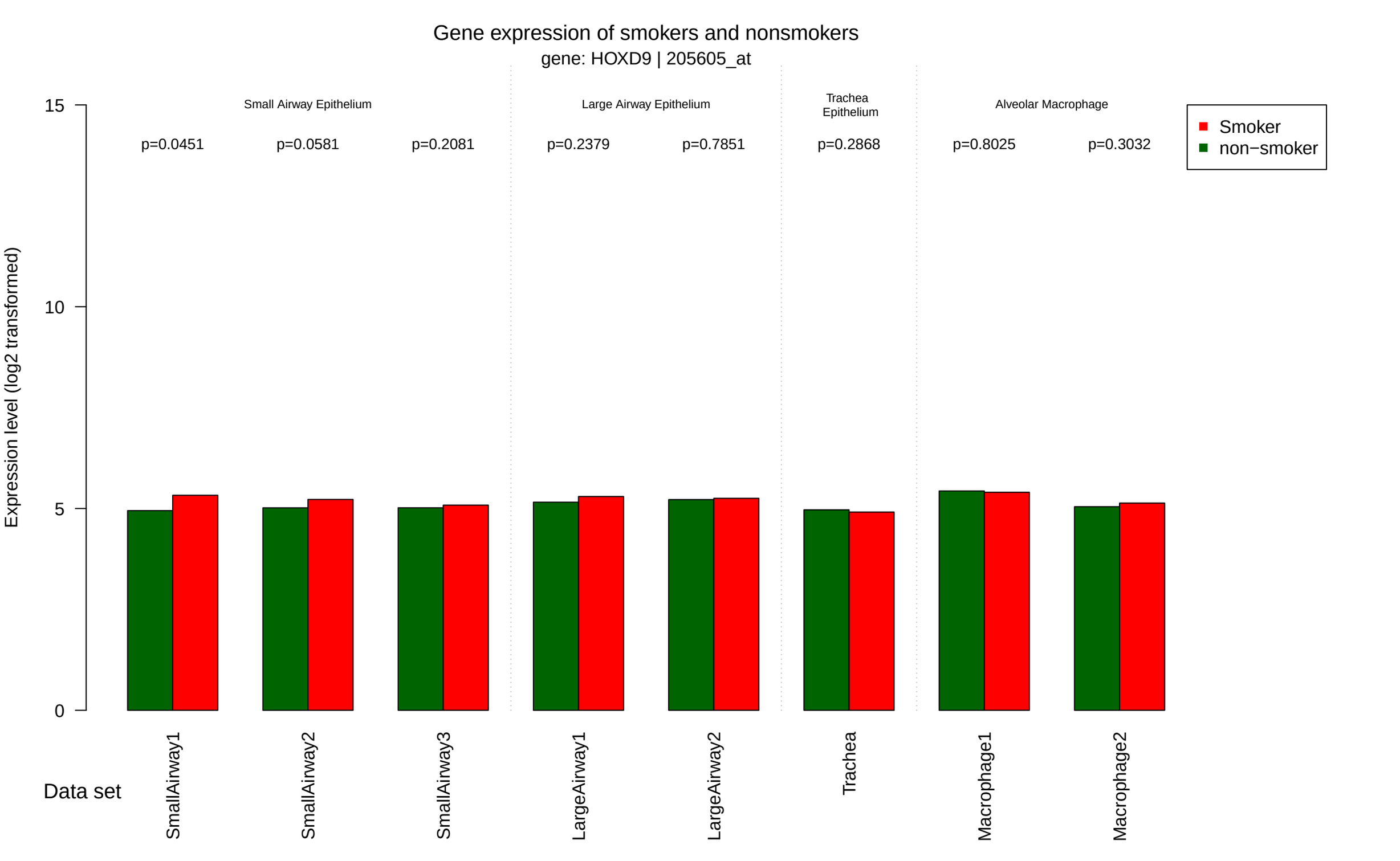
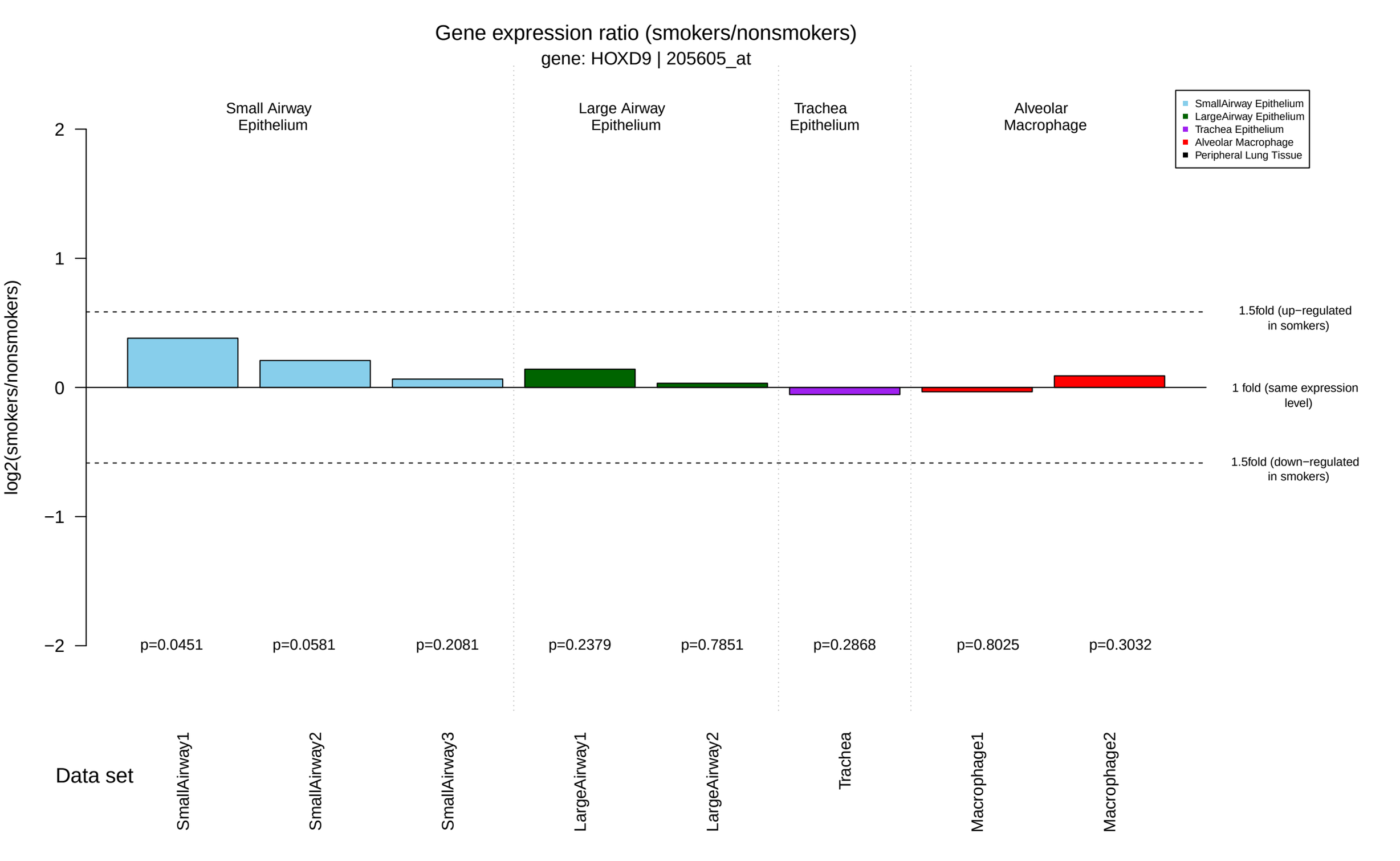
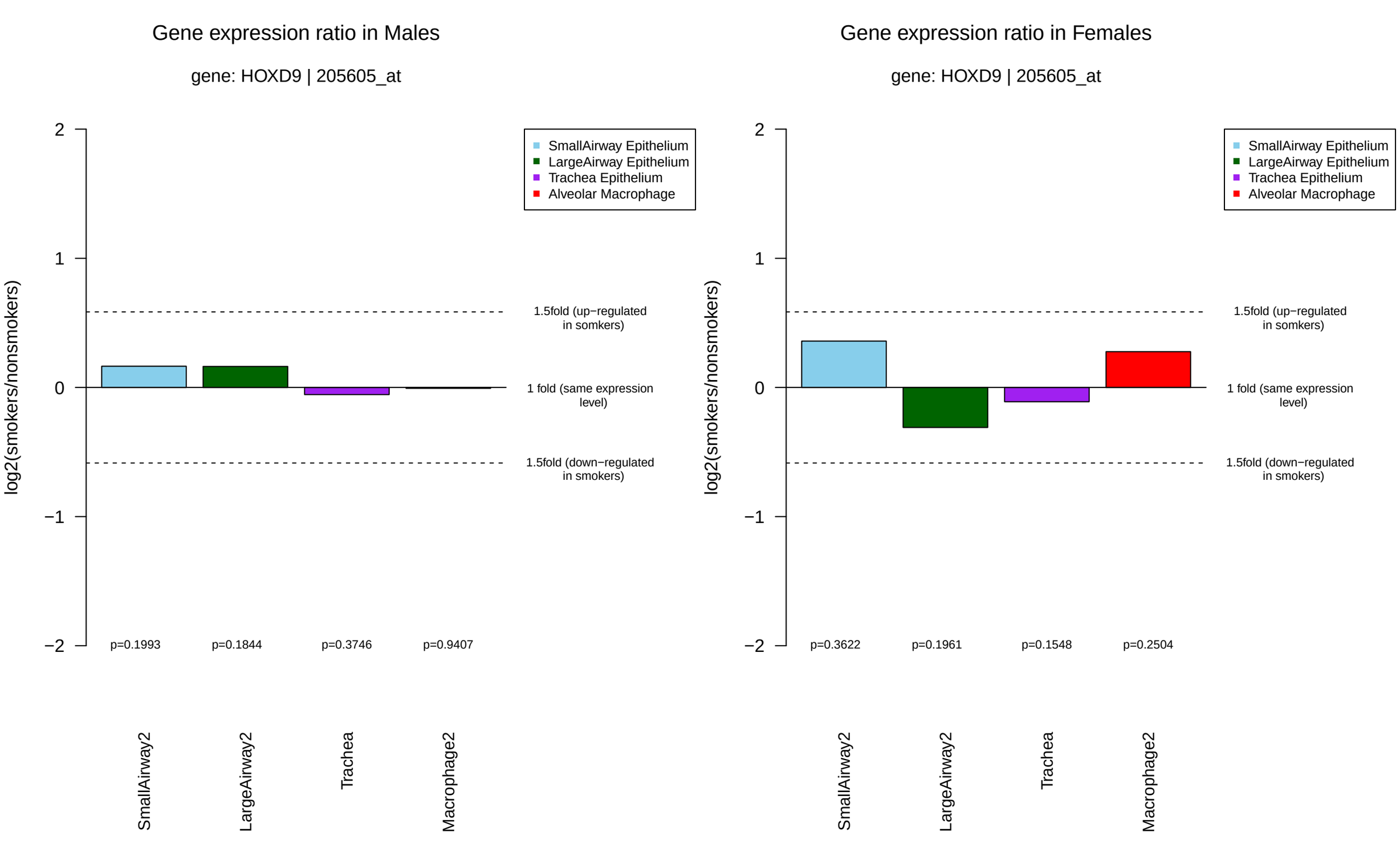
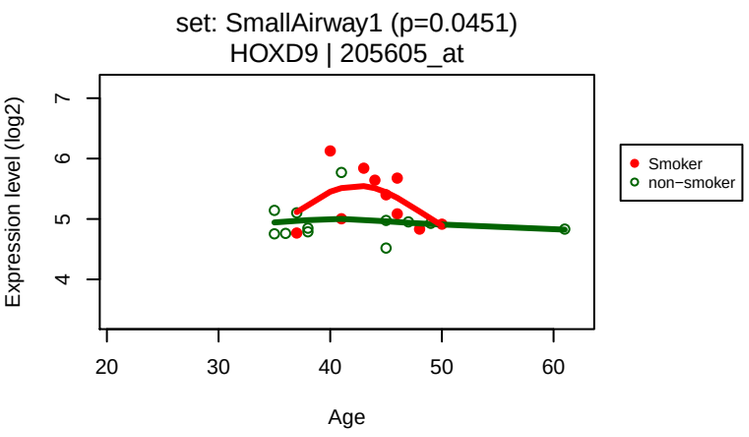
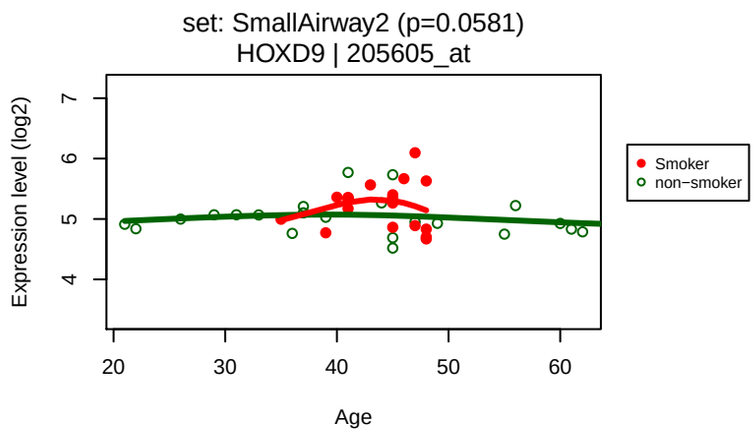
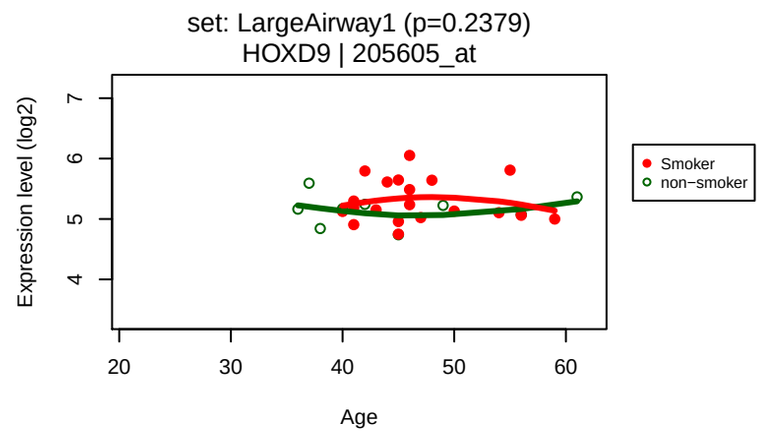

Gene expression of smokers and nonsmokers
gene: HOXD9 | 205605_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0451
p=0.0581
p=0.2081
p=0.2379
p=0.7851
p=0.2868
p=0.8025
p=0.3032
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

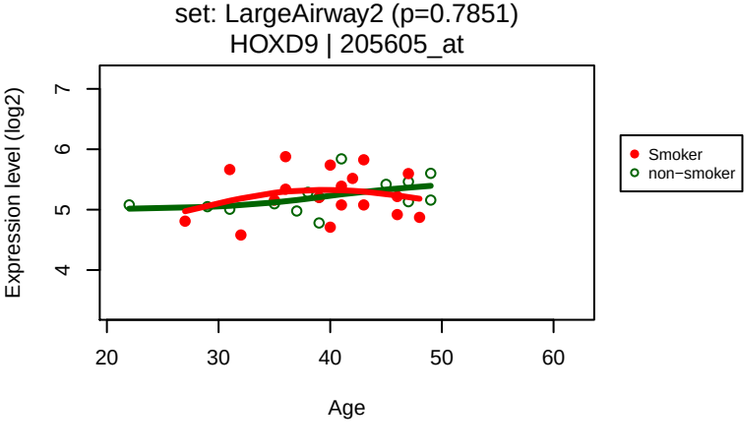

Gene expression ratio (smokers/nonsmokers)
gene: HOXD9 | 205605_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0451
p=0.0581
p=0.2081
p=0.2379
p=0.7851
p=0.2868
p=0.8025
p=0.3032
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

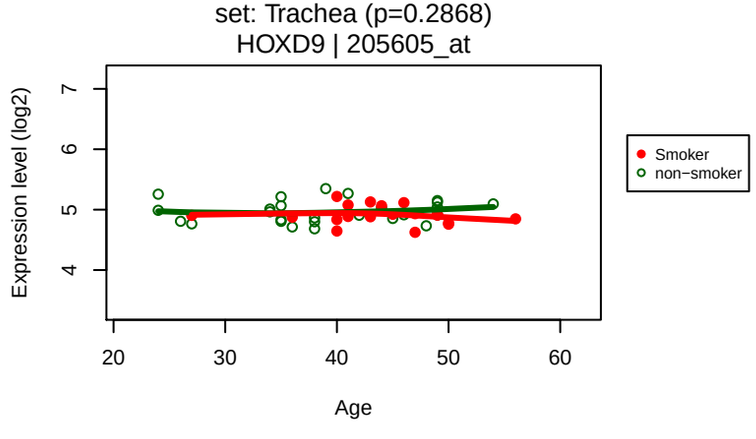

Gene expression ratio in Males
gene: HOXD9 | 205605_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1993
p=0.1844
p=0.3746
p=0.9407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD9 | 205605_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3622
p=0.1961
p=0.1548
p=0.2504
SmallAirway2
LargeAirway2
Trachea
Macrophage2

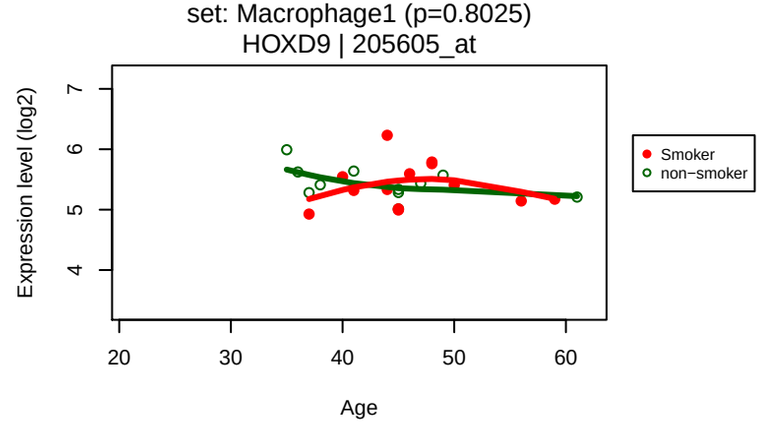

set: SmallAirway1 (p=0.0451)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker

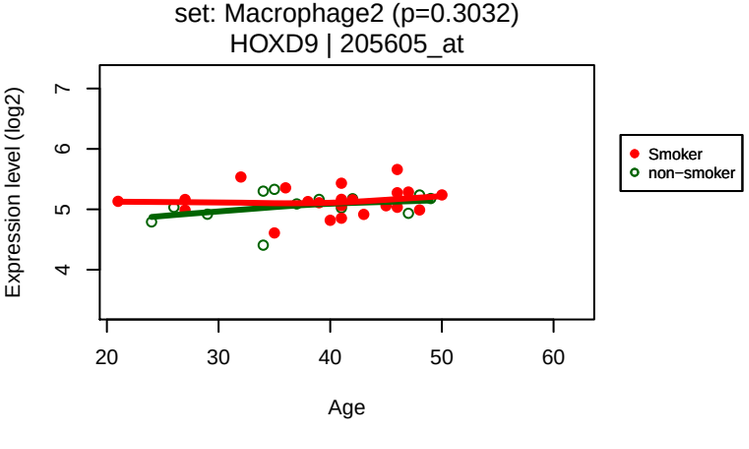

set: SmallAirway2 (p=0.0581)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker

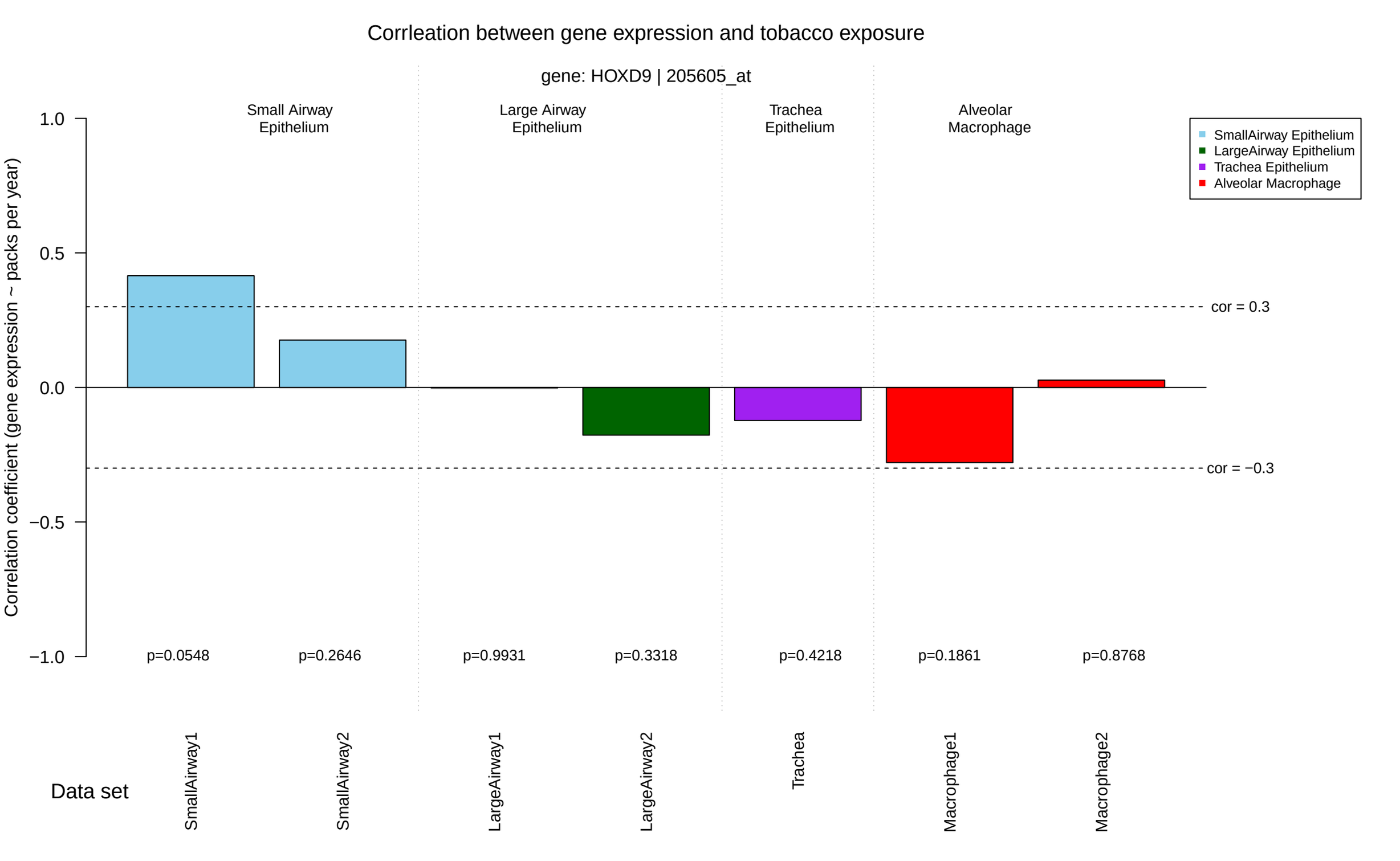

set: LargeAirway1 (p=0.2379)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7851)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2868)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8025)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3032)
HOXD9 | 205605_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD9 | 205605_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0548
p=0.2646
p=0.9931
p=0.3318
p=0.4218
p=0.1861
p=0.8768
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2